Test sheet: the rule for symmetric above marks, and the vertical evidence

Proposed rule. A symmetric above mark (circumflex, tilde, macron, breve, dot above, dieresis, ring, caron) has its anchor on its own axis: $k = K$, where k is the centre of the mark's outline on its axis minus its anchor X, and K is 0 (150 in semibold italic). It is applied only where $|k - K| \geq 10$; smaller deviations are within the designer's own spread. Asymmetric marks (grave, acute, double acute, double grave) keep their designed k. In the patched fonts this moves four mark anchors:

- Italic ◌̃ U+0303: $k = 11$; anchor $(-17, 697) \to (-6, 697)$, so the mark moves 11 left.
- Semibold ◌̈ U+0308: $k = 12$; anchor $(-198, 653) \to (-186, 653)$, so the mark moves 12 left.
- Semibold italic ◌̃ U+0303: $k = 139$; anchor $(-134, 700) \to (-145, 700)$, so the mark moves 11 right.
- Semibold italic ◌̈ U+0308: $k = 112$; anchor $(-210, 645) \to (-248, 645)$, so the mark moves 38 right.

The "rule" fonts below are the patched fonts with only these anchors changed.

1. Horizontal: composed minus precomposed, now and under the rule

Δx: median over the lowercase precomposed letters of the mark's offset, composed (base + CGJ + mark) minus the designer's precomposed glyph, along the slant; positive means the composed mark lands right of the designer's. Under the rule Δx is no longer zeroed per mark: what remains is common to all the marks of a style (about $+12$ to $+16$ in semibold italic, where the designer builds the precomposed glyphs with the base about 12 right), which is a question for the base anchors, not the marks.

	Regular			Italic			Semibold			Semibold italic		
	k	now	rule	k	now	rule	k	now	rule	k	now	rule
◌̂ circumflex	-1	+3	+3	0	+18	+18	0	+0	+0	153	+16	+16
◌̃ tilde	0	-4	-4	11	+24	+13	-4	-9	-9	139	+12	+23
◌̄ macron	-2	-9	-9	-1	+0	+0	6	-2	-2	141	+5	+5
◌̆ breve	0	+8.5	+8.5	0	+5	+5	0	+6	+6	150	+14	+14
◌̇ dot above	0	+3	+3	3	+15	+15	1	-4	-4	148	-8	-8
◌̈ dieresis	0	-0.5	-0.5	0	-3.5	-3.5	12	+13.5	+1.5	112	-26	+12
◌̊ ring	2	+3	+3	3	+8	+8	0	-0.5	-0.5	158	+24	+24
◌̌ caron	2	+5.5	+5.5	1	+10	+10	0	+0	+0	150	+1	+1

2. Vertical: the evidence of Δy

Δy: the same comparison vertically; positive means the composed mark sits higher than the designer's. Median over the precomposed letters with an x-height base (ascender letters such as ƀ ƌ ɦ ǩ excluded: the designer puts the mark beside the ascender), with the range. On x-height bases the base anchor is at the reference height (645) for every letter, so a Δy that is the same on every base belongs to the mark's anchor: it is evidence for moving that anchor down (positive Δy) or up (negative). A spread across bases (italic macron: -1 on ā ē ḡ, -32 on ī ō ū ȳ) means the designer's precomposed glyphs disagree among themselves. Bold: $|\Delta y| \geq 10$.

	Regular		Italic		Semibold		Semibold italic	
	Δy	range	Δy	range	Δy	range	Δy	range
◌̀ grave	-1	-1..-1	-1	-1..-1	-1	-1..-1	+0	0..1
◌́ acute	+0	0..0	+0	-9..5	-1	-1..-1	+6.5	6..9
◌̂ circumflex	-1	-1..1	-1	-8..1	-4	-4..-2	-1	-1..1
◌̃ tilde	-4	-4..-4	-4	-4..1	+14	14..14	-2	-2..-1
◌̄ macron	+18	18..18	-32	-32..-1	-12	-12..-12	-32	-32..-5
◌̆ breve	-6.5	-10..-4	-6.5	-10..-4	-9.5	-13..-7	-7	-10..-4
◌̇ dot above	-45	-46..-44	-11	-15..-10	-21	-23..-20	+11	3..12
◌̈ dieresis	-2	-2..0	+0	0..2	-6	-6..-2	+6	6..10
◌̊ ring	+7.5	7..8	+7	-3..8	+0.5	0..1	+23	16..23
◌̋ double acute	+11	11..11	+11	11..11	-8	-8..-8	+11	11..11
◌̌ caron	+0	-7..2	+0	-7..2	+0	-7..2	+0	-7..2

3. The rule: the four marks it moves

Red: the designer's precomposed glyph. Blue: composed now (upper row) and under the rule (lower row). Purple where they agree.

Italic tilde

now ã ẽ ĩ õ ũ ñ ỹ ṽ

rule ã ẽ ĩ õ ũ ñ ỹ ṽ

Semibold dieresis

now ä ë ï ö ü ÿ ẅ ẍ

rule ä ë ï ö ü ÿ ẅ ẍ

Semibold italic tilde

now ã ẽ ĩ õ ñ ỹ ṽ

rule ã ẽ ĩ õ ñ ỹ ṽ

Semibold italic dieresis

now ä ë ï ö ÿ ẅ ẍ

rule ä ë ï ö ÿ ẅ ẍ

Against the circumflex on the same base. Red: the moved mark; blue: the circumflex, both composed. Upper row now, lower row under the rule.

Italic tilde, now ô ñ î j̃ û â ə̂ ɔ̂

rule ô ñ î j̃ û â ə̂ ɔ̂

Semibold dieresis, now ô ñ î j̈ û â ə̂ ɔ̂

rule ô ñ î j̈ û â ə̂ ɔ̂

Semibold italic tilde, now ô ñ î j̃ û â ə̂ ɔ̂

rule ô ñ î j̃ û â ə̂ ɔ̂

Semibold italic dieresis, now ô ñ î j̈ û â ə̂ ɔ̂

rule ô ñ î j̈ û â ə̂ ɔ̂

In text. Nasal and centralized vowels: precomposed letters (ã ẽ õ ũ, ä ë ö ü) beside composed ones with no precomposed form (ɛ̃ ə̃ ɔ̃ ʊ̃, ë ə̈ ɔ̈ ʊ̈).

Italic, now [ã ɐ̃ ẽ ɛ̃ ə̃ ĩ ɪ̃ õ ɔ̃ ũ ʊ̃ ỹ ʏ̃] [ä ɐ̈ ë ɛ̈ ə̈ ï ɪ̈ ö ɔ̈ ü ʊ̈ ÿ ʏ̈]

rule [ã ɐ̃ ẽ ɛ̃ ə̃ ĩ ɪ̃ õ ɔ̃ ũ ʊ̃ ỹ ʏ̃] [ä ɐ̈ ë ɛ̈ ə̈ ï ɪ̈ ö ɔ̈ ü ʊ̈ ÿ ʏ̈]

Semibold, now	[ã ẽ ẽ ɛ̃ ə̃ ĩ ĩ õ ɔ̃ ũ ʊ̃ ỹ ɤ̃]	[ä ë ë ɛ̈ ə̈ ï ï ö ɔ̈ ü ʊ̈ ÿ ɤ̈]
rule	[ã ẽ ẽ ɛ̃ ə̃ ĩ ĩ õ ɔ̃ ũ ʊ̃ ỹ ɤ̃]	[ä ë ë ɛ̈ ə̈ ï ï ö ɔ̈ ü ʊ̈ ÿ ɤ̈]
Semibold italic, now	*[ã ẽ ẽ ɛ̃ ə̃ ĩ ĩ õ ɔ̃ ũ ʊ̃ ỹ ɤ̃]*	*[ä ë ë ɛ̈ ə̈ ï ï ö ɔ̈ ü ʊ̈ ÿ ɤ̈]*
rule	*[ã ẽ ẽ ɛ̃ ə̃ ĩ ĩ õ ɔ̃ ũ ʊ̃ ỹ ɤ̃]*	*[ä ë ë ɛ̈ ə̈ ï ï ö ɔ̈ ü ʊ̈ ÿ ɤ̈]*

4. How much the marks dance (patched fonts, now)

Composed (blue) over precomposed (red), on o. Each pair is the designer's precomposed letter and o + CGJ + the same mark. A vertical offset is the Δy of section 2, a horizontal one the Δx.

Regular 

Italic 

Semibold 

Semibold italic 

In a line. Each pair is a precomposed letter followed by a composed one with no precomposed form, on a base of the same height: o with ɔ, e with ə, u with ʊ. In the composed letters the marks are placed by the anchors; in the precomposed letters, where the designer drew them. A mark that jumps between the two letters of a pair is a mark whose Δy (or Δx) is large.

Regular

òɔ́ ôɔ̃ ōɔ̄ ǒɔ̌ ȯɔ̇ öɔ̈ ő̆ ő̆

èə́ êə̃ ēə̄ ěə̌ ėə̇ ëə̈

ùʊ́ ûʊ̃ ūʊ̄ ǔʊ̌ üʊ̈ ůʊ̊ ű̆ ű̆ ǔʊ̌

Italic

òɔ́ ôɔ̃ ōɔ̄ ǒɔ̌ ȯɔ̇ öɔ̈ ő̆ ő̆

èə́ êə̃ ēə̄ ěə̌ ėə̇ ëə̈

ùʊ́ ûʊ̃ ūʊ̄ ǔʊ̌ üʊ̈ ůʊ̊ ű̆ ǔʊ̌

Semibold

òɔ́ ôɔ̃ ōɔ̄ ǒɔ̌ ȯɔ̇ öɔ̈ ő̆ ő̆

èə́ êə̃ ēə̄ ěə̌ ėə̇ ëə̈

ùʊ́ ûʊ̃ ūʊ̄ ǔʊ̌ üʊ̈ ůʊ̊ ű̆ ǔʊ̌

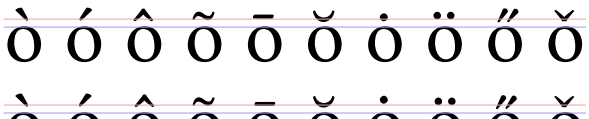
Semibold italic

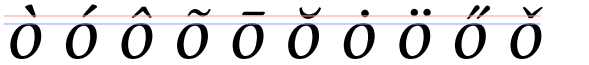
òɔ́ ôɔ̃ ōɔ̄ ǒɔ̌ ȯɔ̇ öɔ̈ ő̆ ő̆

èə́ êə̃ ēə̄ ěə̌ ėə̇ ëə̈

ùʊ́ ûʊ̃ ūʊ̄ ǔʊ̌ üʊ̈ ůʊ̊ ű̆ ǔʊ̌

One base, every mark. o + CGJ + each mark (upper row) and the designer's ò ó ô õ ō ǒ ȯ ö ő̆ ǒ (lower row), with lines at the x-height and 100 above it.

Regular 

Italic 

ò ó ô õ ō ŏ ȯ ö ő ǒ

Semibold

ò ó ô õ ō ŏ ȯ ö ő ǒ

ò ó ô õ ō ŏ ȯ ö ő ǒ

Semibold italic

ò ó ô õ ō ŏ ȯ ö ő ǒ

ò ó ô õ ō ŏ ȯ ö ő ǒ